Case1

1. 初始剛度（Initial Stiffness）與彈性階段之落差

由曲線初期可見，OpenSees 的斜率明顯低於 XTRACT。

- **OpenSees 邏輯：** 程式碼中採用 Concrete01 模型，其初始切線剛度 E_c 是由 $2\times\frac{f'_c}{\varepsilon_{c0}}$ 定義。由於設定 $\varepsilon_{c0}=0.002$，這使得模型在抵達峰值前的路徑較為平緩。
- **XTRACT 邏輯：** 該軟體通常採用規範公式（如 ACI 318 的 $4700\times\sqrt{f'_c}$）來定義混凝土模數，其數值通常大於纖維模型計算出的割線剛度，導致其曲線在初期展現出更高的剛性。

2. 保護層剝落（Cover Spalling）與局部強度波動

曲線在應變約 0.002 處出現明顯的分歧，OpenSees 橘線出現了一次明顯的「凹陷」。

- **OpenSees 的細緻模擬：** 在腳本中，截面被明確劃分為核心混凝土與保護層混凝土。保護層模型設定為在應變達到 0.004 時強度降至零（$f'_{cu}=0$）。當分析進行至此區間時，保護層纖維失效，造成總承載力的暫時性下降，這真實地模擬了 RC 構件在受壓時的「保護層剝落（Spalling）」現象。
- **XTRACT 的平滑處理：** XTRACT 在截面整合運算時，往往對保護層失效過程進行了演算法上的平滑（Smoothing），或其採用的約束模型對於保護層損壞的權重設定不同，因此曲線較為連續。

3. 峰值強度與圍束效應（Confinement Effect）

兩者的最高承載力存在約 12% 的偏差（XTRACT 約 16,000 kN；OpenSees 約 14,000 kN）。

- **參數對應問題：** 雖然 OpenSees 計算了 Mander 圍束參數，但 Concrete01 屬於線性軟化模型（Linear Softening），其在峰值後的能量消散描述與 XTRACT 內建的 Mander 數學公式（通常包含更複雜的非線性硬化段）存在定義上的落差，這導致了兩者對於「圍束帶來的強度增益」預估不一致。

4. 後期硬化行為（Post-peak Hardening）

在應變超過 0.004 後，OpenSees 曲線呈現緩步爬升趨勢，而 XTRACT 則趨於平緩。

- **鋼筋模型影響：** 關鍵在於 OpenSees 採用了 ReinforcingSteel 材料卡，該模型考慮了鋼筋的應變硬化（Strain Hardening）性質（硬化係數 0.018）。
- **分析結論：** 由於本案配置了 25 支 D25 縱向鋼筋，當應變進入較大範圍時，鋼筋進入硬化段所產生的強度增益足以抵銷混凝土壓碎造成的損失，使得構件後期展現出韌性。反觀 XTRACT 若採用理想彈塑性模型（Elastic-Perfectly Plastic），則無法捕捉到此硬化段的行為。

總結

Case 1 的比較說明： OpenSees 雖然在初始剛度的預估上較為保守，但其透過纖維劃分與精確的 ReinforcingSteel 參數，成功捕捉到了「保護層剝落」與「鋼筋硬化」這兩個關鍵的物理現象。XTRACT 則在預測構件初期剛度與穩定承載力方面較具優勢，但在非線性行為的靈敏度上略遜於 OpenSees。

Case 1

1. 開場與剛度對比 (Introduction & Stiffness)

"First, let's look at the beginning of the curve. You'll notice that **XTRACT** is much steeper than **OpenSees** model. This comes down to how each software defines 'stiffness'. **XTRACT** usually implement standard ACI formulas like 4700×the square root of f'c, which is pretty stiff. But in **OpenSees** code, we used the Concrete01 material with a peak strain $\varepsilon_{c0} = 0.002$. This makes the initial curve a bit 'softer' or more flexible compared to the code-based formulas."

2. 神秘的「凹陷」與保護層剝落 (The "Dip" & Cover Spalling)

"Now, here's a little 'dip' in the orange line around strain **0.002**. **XTRACT** doesn't show this, but **OpenSees** does. In our OpenSees script, we split the column section into **core** and **cover** concrete. What you're seeing there is the concrete **cover spalling**. Once the strain hits **0.004**, the cover concrete basically loses its strength. **OpenSees** is sensitive enough to capture this sudden drop in capacity, while **XTRACT** just smooths the whole thing over."

3. 峰值強度與圍束效應 (Peak Strength & Confinement)

"Regarding the peak strength, **XTRACT** predicts a higher capacity—about **16,000 kN** compared to OpenSees **14,000 kN**. This is likely because **XTRACT**'s internal math for confinement is more 'aggressive' than the linear softening path in Concrete01 in OpenSees."

4. 後期的爬升行為 (Post-peak Hardening)

"Finally, look at the end of the curve. After the peak, **OpenSees** actually starts climbing again while **XTRACT** stays flat. This is because we chose the ReinforcingSteel material, which includes **strain hardening**. Since this column has **16-D25 bars**. Once the concrete starts failing, the steel takes over and keeps the strength rising as the strain increases. It shows that the **OpenSees** fiber model is much better at simulating these complex, late-stage behaviors."

Case 2

A. 延續 Case 1 的模型特徵 (Consistency)

- **初始剛度偏差**：如同 Case 1，OpenSees 的初始斜率較 XTRACT 平緩。這驗證了在相同混凝土強度（28MPa）下，Concrete01 的參數設定（$\varepsilon_{c0} = 0.002$）穩定地產生了較保守的剛度預估。
- **保護層剝落 (Cover Spalling)**：在應變約 **0.004** 處，OpenSees 再次精準捕捉到強度凹陷。這對應了程式碼中對保護層失效的明確定義（f'cu = 0），再次突顯 OpenSees 纖維模型在模擬物理剝落上的優勢，而 XTRACT 則維持平滑曲線。

B. SD550 帶來的演進差異 (Evolution)

- **強勁的後期爬升**：相較於 Case 1，Case 2 在應變 0.004 之後的強度恢復更為顯著，甚至在 0.008 處推回了一個明顯的**二次峰值**（約 15,000 kN）。
- **鋼筋硬化敏感度**：由於使用了 **SD550**（f_y = 550 MPa）配合 ReinforcingSteel 材料模型，高強度鋼筋在進入應變硬化段後，提供的強度增益遠大於 Case 1，這解釋了為何 OpenSees 在大應變下的行為會與 XTRACT 的理想彈塑性假設拉開更大的差距。

Case 2

"Moving on to **Case 2**, we should view this result as a **direct evolution of Case 1**.

First, look at the **initial stiffness**. Just like the previous case, the **OpenSees** curve is flatter than **XTRACT**. Since the concrete is still 28MPa, our OpenSees model consistently uses the same setting, which leads to a softer initial response compared to the standard formulas in **XTRACT**.

The most consistent part is that **'dip' at 0.004 strain**. Because fiber section and material definitions for the cover concrete haven't changed, **OpenSees** once again captures the **cover spalling**. It proves that while **XTRACT** gives us a smooth, idealized curve, **OpenSees** is showing us the actual physical failure of the column.

The real 'plot twist' here is the **SD550 steel**. After the cover spalling, the **OpenSees** curve takes a very aggressive climb back up, creating a massive **second peak** at around 0.008 strain. Because we upgraded the steel grade, the rebars enter the **hardening stage** with much more strength. Once the high-strength rebar enters the strain hardening stage, it gives a much bigger strength boost than in Case 1. That's why at large strains, OpenSees shows a huge difference compared to XTRACT's simple elastic-plastic model.

Comparing Case 1 and Case 2, the most significant impact of upgrading the reinforcement from **SD420 to SD550** is the dramatic improvement in the column's strength recovery after the cover concrete peels off. While both models exhibit a similar 'dip' at **0.004 strain** due to the defined cover spalling, the higher yield strength of **SD550** allows the longitudinal bars to enter the **strain hardening** region with much more force. This extra resistance effectively compensates for the loss of the concrete shell, creating a very prominent **'second peak'** in Case 2 that we don't see as clearly in Case 1. Ultimately, increasing the steel grade doesn't just bump up the peak capacity; it significantly enhances the member's resilience and ductility at large strains by leveraging the hardening properties of the ReinforcingSteel model."

Case 3

初始剛度的一致性提升：

- **現象**：不同於 28MPa 的案例，Case 3 中 OpenSees 與 XTRACT 的初始斜率（剛度）非常接近。
- **原因**：在 OpenSees 程式碼中，E_c是由 $2\times\frac{f'_c}{\varepsilon_{c0}}$ 計算。當 f'c 提升至 70MPa 且ε_{c0}維持 0.002 時，計算出的剛度高達 70,000 MPa。這使得 OpenSees 的模型剛度趕上了 XTRACT 的標準公式，兩者在彈性階段高度重合。

脆性剝落與明顯的強度驟降 (Brittle Spalling)：

- **現象**：OpenSees 在應變 **0.004** 處出現了一個非常陡峭且深陷的強度降幅（Dip）。
- **原因**：高強度混凝土極為脆性。在程式碼中，保護層在應變 0.004 時強度直接歸零。因為 70MPa 混凝土所分擔的軸力極大，一旦失效，對總承載力的打擊比 28MPa 案例顯著得多，導致了圖中劇烈的波動。

後期下降段的斜率差異：

- **現象**：在 0.005 應變之後，OpenSees 的下降斜率比 XTRACT 更陡。
- **原因**：OpenSees 使用的 Concrete01 是**線性軟化模型（Linear Softening）**，在達到峰值後會以直線路徑下降。而 XTRACT 採用的模型（通常為 Mander 改良型）在處理高強度混凝土時，其曲線下降段較具韌性且呈非線性，因此 XTRACT 在大應變下的殘餘強度預估較高。

Case 3

"Now we're looking at **Case 3**, where things get a bit more intense because we've switched to **70MPa High-Strength Concrete (HSC)**.

First, check out the **initial stiffness**. Unlike the first two cases, the **OpenSees** and **XTRACT** curves are practically on top of each other at the start. This is because 70MPa is so stiff that the formula in OpenSees code catches up to the high stiffness values **XTRACT** uses. So you can see that the initial stiffness of OpenSees and Xtract are similar.

But here is where it becomes different: the **'dip' at 0.004 strain**. In this case, the drop in **OpenSees** is much sharper and deeper than before. Since 70MPa concrete is very brittle, losing that cover layer is a huge problem. Our OpenSees script kills the cover strength at exactly 0.004, and because that cover was carrying a massive load, the strength drop quickly after the cover concrete spalling.

Overall, **Case 3** proves that as concrete gets stronger, the timing and severity of **cover spalling** become the most critical parts of the simulation."

Case 4

初始剛度的高度重合 (Stiffness Convergence)

- **現象**：與 Case 3 相同，OpenSees 與 XTRACT 的初始斜率幾乎完全疊合。
- **原因**：在 70MPa 的高壓應力下，OpenSees 腳本中計算出的 E_c（約 70,000 MPa）與 XTRACT 的規範公式非常接近。這證明在模擬高強度材料時，兩軟體的彈性階段預估具備很高的一致性。

保護層剝落造成的劇烈衝擊 (The Spalling Dip)

- **現象**：OpenSees 在應變 **0.004** 處出現了極為顯著的強度驟降（Dip），其跌幅深度較 28MPa 案例更為驚人。
- **原因**：這是高強度混凝土「脆性」的體現。在程式碼中，保護層混凝土（IDUC）在應變達到 0.004 時強度歸零。由於 70MPa 混凝土承載比重極高，一旦剝落，構件會瞬間喪失巨大承載力，反映出 OpenSees 對於物理失效邊界的敏感模擬。

SD550 鋼筋帶來的「二次高峰」 (The Second Peak)

- **現象**：在 0.004 驟降後，OpenSees 曲線迅速反彈，並在應變約 **0.005** 處推回另一個波峰。
- **原因**：這是 **SD550** 高強度鋼筋的作用。雖然保護層剝落了，但由於縱向鋼筋擁有極高的降伏強度與硬化增益（$f_y=550$ MPa），它在大應變下提供的抗力足以抵銷部分混凝土的損失，展現出構件在破壞後期的韌性潛力。

後期下降段的模擬落差

- **現象**：在 0.006 之後，OpenSees 的曲線呈線性陡降，而 XTRACT 則顯得較為平滑穩定。
- **原因**：這再次凸顯了 Concrete01 **線性軟化模型**的局限性。XTRACT 的內建 Mander 模型對高強度混凝土的圍束行為有更圓滑的非線性描述，導致其在極大應變下的殘餘強度預估較高。

Case 4

"Finally, we have **Case 4**, the ultimate combination of **70MPa concrete** and **SD550 steel**.

"Building on what we saw in **Case 3**, **Case 4** shows us how high-strength steel can actually 'save' a column that has such brittle concrete.

As expected, the **initial stiffness** is still identical between **OpenSees** and **XTRACT**, confirming that at **70MPa**, the concrete stiffness dominates the elastic range. But the real story starts after the **spalling dip at 0.004 strain**.

In Case 3, the column struggled to recover, but here in **Case 4**, the upgrade to **SD550 steel** makes a difference. Because the yield strength is higher, those **SD550 rebars** enter the **hardening stage** with enough power to fight back against the loss of the concrete cover. This creates a much more significant **second peak** compared to what we saw in the previous case.

So, while **Case 3** highlighted the **brittleness** of the material, **Case 4** demonstrates the **resilience** provided by high-strength steel SD550. It proves that **OpenSees** is sensitive enough to show how **SD550** reinforcement acts as a safety mechanism, sustaining the load even after the 70MPa concrete fails."

Conclusions

In summary, while XTRACT provides a stable and efficient estimate of ultimate capacity through numerical smoothing and stiffness formulations, OpenSees offers a much more sensitive look into the physical failure mechanisms of reinforced concrete columns. By using fiber discretization, OpenSees successfully captures the "spalling dip" at 0.004 strain—a phenomenon that becomes significantly more brittle and dramatic in our high-strength concrete cases—while also modeling the aggressive strength recovery and "second peak strength" triggered by the strain hardening of our SD550 reinforcement.

Ultimately, this comparison demonstrates that while XTRACT is ideal for rapid design verification, the OpenSees fiber model is suitable for research-level simulations because it easily accounts for localized material failure and the late-stage hardening benefits of high-performance steel that idealized elastic-plastic models tend to overlook.